What's New in Voltus Stylus Common UI

Product Version 26.10
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Contents

About This Manual

This manual provides information about Product Version 26.1 of the Cadence® Voltus™ IC Power Integrity Solution suite, also known as Voltus. The Voltus family encompasses the following products:

- Voltus IC Power Integrity Solution - L
- Voltus IC Power Integrity Solution - XL

Related Documents

For more information about Voltus, see the following documents. You can access these and other Cadence documents using the Cadence Help documentation system.

Voltus IC Power Integrity Solution Product Documentation

- *Voltus Known Problems and Solutions*
  Describes important Cadence Change Requests (CCRs) for Voltus, including solutions for working around known problems.
- *Voltus Text Command Reference*
  Describes the Voltus text commands, including syntax and examples.
- *Voltus Menu Reference*
  Provides information specific to the forms and commands available from the Voltus graphical user interface.
- *Voltus User Guide*
  Provides information on using various Voltus features.
- README file
  Contains installation, compatibility, and other prerequisite information, including a list of Cadence Change Requests (CCRs) that were resolved in this release. You can read this file online at downloads.cadence.com.

Voltus Stylus Common UI Product Documentation

- *Voltus Stylus Common UI Text Reference Manual*
  Describes the Voltus Stylus Common UI text commands, including syntax and examples.
- *Voltus Stylus Common UI User Guide*
  Provides information on using various Voltus Stylus Common UI features.

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Stylus Release Overview

Release 26.10 Overview

Changes to Command Options and Defaults

New Command Parameters

The following table lists the command parameters that were added to the Voltus IC Power Integrity software. The second column identifies the corresponding chapter in *Voltus Text Command Reference* where the command is documented.

New Parameters	Chapter
`check_pg_library` `-libgen_command_file`	Power-Grid Library Commands

`get_ir_diagnostics`
`-cap_lower_limit`
`-cap_upper_limit`
`-report_mode`
	Rail Analysis Commands

`get_voltus_diagnostics_mode` `-timing_protection_config`	Rail Analysis Commands
`read_power_rail_results` `-force`	GUI Commands

report_power -cell_fit_file -output_cell_fit_file	Power Calculation Commands and Attributes
report_self_heat -parameter_file	Rail Analysis Commands
set_advanced_pg_library_mode -pgv_path -reuse_electrical_data -spectre_standalone_simulation -use_physical_view_from_techpgv	Power-Grid Library Commands
set_die_model -static_current	Rail Analysis Commands
set_power_data -cell -power_directory	Rail Analysis Commands
set_pg_library_mode -tsv_subckt_modelfile_list	Power-Grid Library Commands
set_rail_analysis_config -inst_ignore_file -lifetime_rating_factor_file -report_compress_level -single_partition_extraction_layer - single_partition_extraction_layer_list -xp_reuse_hpt_directory -xp_reuse_parse_def_directory	Rail Analysis Commands
set_self_heat_analysis_mode -parameter_file	Rail Analysis Commands

set_thermal_analysis_mode -thermal_output_dir	Power Calculation Commands and Attributes
set_voltus_diagnostics_mode -all_scenario_ranking -disable_layer -ground_reff_threshold -ground_rlrp_threshold -initialize -input_insight_db -instance_mapping -master_list -master_mapping -output_insight_db -path_group -power_reff_threshold -power_rlrp_threshold -scenario_ranking -timing_protection_config -violation_input	Rail Analysis Commands
write_current_region -on_interdie_bump	Rail Analysis Commands
write_die_model -local_reference_ground -suppress_negative_rc	Rail Analysis Commands

New Root Attributes

The following table lists the root attributes that were added to the Voltus Stylus Common UI software. The second column identifies the category in which the attribute has been added and the third column identifies the corresponding chapter of the *Text Command Reference* where the root attribute is documented.

New Root Attributes	Category	Chapter

`check_ac_limit_lifetime_rating_factor_file` `check_ac_limit_report_unannotated_shapes`	`check_ac_limit`	Check Commands and Attributes

`power_effective_load_cap_slew_threshold_limit`
`power_enable_force_propagation`
`power_ignore_unconstraint_net_with_timing_data`
`power_off_pg_bulk_leakage`
`power_partition_spef`
`power_show_cell_usage_stats`
`power_use_effective_load_cap_method`
	`power`	Power Calculation Commands and Attributes

Obsolete Command Parameters

The following obsolete text command parameters have been removed from the software.

`set_advanced_pg_library_mode`

`-import_xdspf_list_file`

`-pgdb_layermap`

`-pgdb_list_file`

`-xdspf_layermap`

These parameters have not been replaced.

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Licensing

Release 26.10 Enhancements

New Voltus Product Option

This release introduces the following product option:

Name	Short Name	Product Number	TCL Option	Description

<table>
<tr>
<td>Voltus InsightAI Multi-Dimensional Option</td>
<td>Voltus InsightAI MD Opt</td>
<td>VTS208</td>
<td>vtsvimd</td>
<td>This option enables the following advanced IR features within Innovus:

Multi-Dimension fixing in a single run by honoring the impact of multiple vectors, multiple PVTs, multiple EIV measurements, and multiple domains with varying thresholds
Hierarchical context in block-level IR fixing including multiple instantiations
Advanced reporting and debugging

VTS208 is an additional option required on top of INVS58. The licenses required to enable the advanced IR fixing features:
INVS 100 (Innvous Base) + INVS Technology Node Option + INVS58 + VTS208 (1 or more depending on usage)</td>
</tr>
</table>

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Library Generation and Extraction

Release 26.10 Enhancements

New Parameter Introduced for Standard Cell PGV Verification

A new parameter, `-spectre_standalone_simulation`, has been added to the `set_advanced_pg_library_mode` command to support verification of power-gate parameters (Idsat, Ileakage, and Ron) and coupled capacitance in Spectre. Earlier, the Spectre setup did not contain computation equations (that is, `.measure` statements). Now, when the `-spectre_standalone_simulation` parameter is set to `true`, the generated setup file includes `.measure` statements. You can use this setup file directly in Spectre to understand and verify the underlying computational processes.

The table below displays a snippet of the computation equations extracted from the .sp file:

CoupledCap1.sp

```
.measure ac CLKINVX20___VDD___IReal find IR(vdd_x0) at=cap_frequency
.measure ac CLKINVX20___VDD___IImg find II(vdd_x0)  at=cap_frequency


*Calculate numerator and denominator of RL model.*
.measure ac CLKINVX20___VDD___Numr param='4*CLKINVX20___VDD___RS*CLKINVX20___VDD___RS*CLKINVX20
___VDD___IImg*CLKINVX20___VDD___IImg'

.measure ac CLKINVX20___VDD___Denom param='2*2*3.14*cap_frequency*CLKINVX20___VDD___RS*CLKINVX20
___VDD___RS*CLKINVX20___VDD___IImg'

.measure ac CLKINVX20___VDD___Numr_estimated param='4*CLKINVX20___VDD___RS_estimated*CLKINVX20
___VDD___RS_estimated*CLKINVX20___VDD___IImg*CLKINVX20___VDD___IImg'

.measure ac CLKINVX20___VDD___Denom_estimated param='2*2*3.14*cap_frequency*CLKINVX20
___VDD___RS_estimated*CLKINVX20___VDD___RS_estimated*CLKINVX20___VDD___IImg'

*Cap will be picked from either of three characterization based on following Rule :
*  1. IF CLKINVX20___VDD___Numr is zero CLKINVX20___VDD__CC won't get picked.*  2. IF CLKINVX20
___VDD___Numr_estimated is zero CLKINVX20___VDD__CC_estimated won't get picked.
*  3. IF (CLKINVX20___VDD___Denom) <= 0 or (CLKINVX20___VDD___Denom) <= 1e-16) --> CLKINVX20___VDD__CC won't get
picked.*  4. IF (CLKINVX20___VDD___Denom_estimated) <= 0 or (CLKINVX20___VDD___Denom_estimated) <= 1e-16) -->
CLKINVX20___VDD__CC_estimated won't get picked.


*series R C with leakage current modeled by RL
.measure ac CLKINVX20___VDD___CC param='(1-sqrt(1-CLKINVX20___VDD___Numr))/(CLKINVX20___VDD___Denom)'

* If X_dIrealLow is zero, above current model will assume X_dIrealLow as 1.
.measure ac CLKINVX20___VDD___CCEstimated param='(1-sqrt(1-CLKINVX20___VDD___Numr_estimated))/(CLKINVX20
___VDD___Denom_estimated)'

*series R C model.
.measure ac CLKINVX20___VDD___CCOld param='(CLKINVX20___VDD___IReal*CLKINVX20___VDD___IReal+CLKINVX20
___VDD___IImg*CLKINVX20___VDD___IImg)/(2*3.14*cap_frequency*CLKINVX20___VDD___IImg)'
```

This parameter is applicable only for standard cell PGV generation (`-cell_type stdcells`).

Support to Specify Additional LibGen Variables

You can now specify the additional LibGen variables using the `-libgen_command_file` parameter of the `check_pg_library` command to verify the consistency between the OBS via geometries and port geometries and ensure that the LEF result aligns with the PGV result.

For example, using the `-libgen_command_file` parameter, you can enable the additional LibGen variables `copy_intersecting_obs_vias_to_ports` and `input_type check_lef_pgv_compatibility`, respectively, to verify the consistency between the generated PGV and the original LEF file.

Use Model:

```
check_pg_library -libgen_command_file filename
```

MIMCap PGV Characterization

The Metal-Insulator-Metal Capacitor (MIMCap) PGV generation flow has been introduced for the simplified modeling of MIMCap cells. Since MIMCaps are inherently parallel-plate structures and do not involve MOSFET devices, this flow enables more accurate and efficient capacitance extraction.

You can generate PGV using either GDS or SPICE inputs for a MIMCap cell:

- **Standard Cell MIMCap PGV Generation**: Only SPICE netlist is considered, and no GDS data is provided for characterizing the MIMcap. The `.param capflag_mim=1` parameter should be specified in the Spice models for MIMCap computation.
- **Macro PGV Generation**: GDS data is used for characterizing MIMcap and an XTC command file is required to specify layer operations to derive the MIMCap shapes.

Ability to Generate New Tech PGV with Updated Tap Node Distance

Now, Voltus allows you to change the tap node distance of LEF based PGVs (standard cells and LEF macros) by only generating a new tech PGV, without regenerating the entire PGV. To support this enhancement, a new option `-use_physical_view_from_techpgv` has been added to the `set_advanced_pg_library_mode` command. The default value of the parameter is `false`.

You can generate a tech PGV with new tap distance as follows:

1. Specify the settings to update the tap node distance through the command line.
   TCL command:
   ```
   set_advanced_pg_library_mode
   -tap_node_distance (With new tap distance)
   -use_physical_view_from_techpgv true
   ```
2. Pass the powergate information if a powergate cell is present.

For example:

```
set_pg_library_mode -power_switch_parameters {…}
```

3. Generate the output for power-grid library generation.

```
write_pg_library -out_dir tech_pgv
```

The generated report will have a note related to "`use_physical_view_in_emir`" flag set as shown in the following snippet:

```
-------------------------------------------------------------------------------
This PGV has "use_physical_view_in_emir" flag set.

Cell               PowerNets       Capacitance     PowerGrid Views
----------------------------------------------------------------------------------------|

DECAP1
                   VDD(0.9000)     5.00000e-15          TECH(12 taps)
                   VSS(0.0000)     5.00000e-15          TECH(12 taps)
 Cell Type = DECOUPLING CAP                                    Cell_Status: PASS
INFO: This is a decap cell.
----------------------------------------------------------------------------------------||
```

You can pass the new generated library with the previously generated PGVs during rail analysis, so that the updated tech PGV is used for tap node distance and electrical parameters (currents, capacitances) come from original PGVs.

Ability to Characterize Single Triangular Current Waveform

Now, the `SINGULAR_TRIANGULAR` syntax is supported in the Datasheet method to characterize a single triangular current waveform for the specified mode. By default, the Datasheet method continues to generate dual peak - a main peak and an additional pre-charge peak.

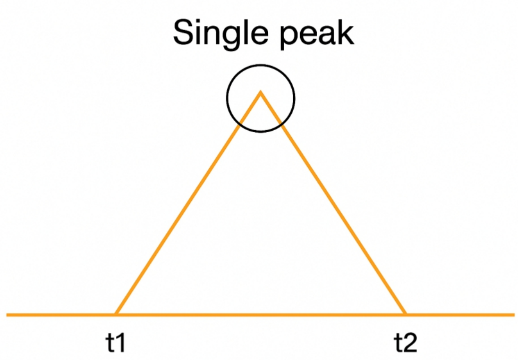
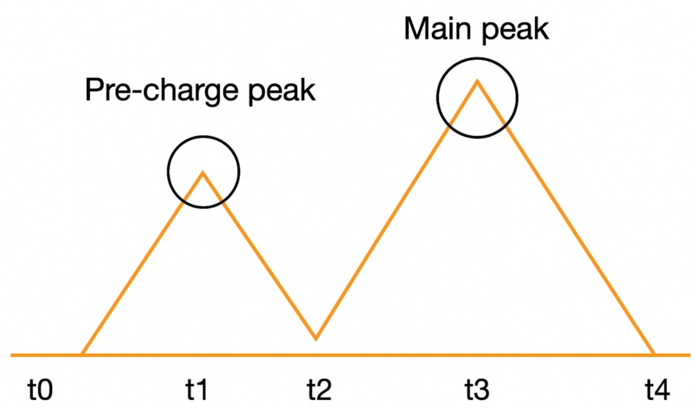
Syntax:

```
SINGLE_TRIANGULAR mode1 mode2
```

Example:

The following is the trigger file format for singular peak and dual peak:

Singular Peak	Dual Peak (default)

CURRENT_CHARACTERIZATION_METHOD DATASHEET CELL ARM_32x64 **SINGLE_TRIANGULAR READ** MODE READ CONDITIONAL_INPUT = (!CEN & !WEN) CONDITIONAL_PIN CLK { RISE } DATASHEET DELAY 1.6879ns DATASHEET_INPUT_SLEW 5.962569e-03ns DATASHEET_OUTPUT_SLEW 5.968207e-03ns DATASHEET_PARAMETER {{ VDDG2 0.1715pf 1.355e-06mW} {VDDG 0.1715pf 3.62e-06mW} {VDD 0.1715pf 2.614e-06mW}} PEAK_CURRENT {{VDDG2 0.25mA {VDDG 0.2mA} {VDDG 0.15mA}} END_MODE	CURRENT_CHARACTERIZATION_METHOD DATASHEET CELL ARM_32x64 MODE READ CONDITIONAL_INPUT = (!CEN & !WEN) CONDITIONAL_PIN CLK { RISE } DATASHEET DELAY 1.6879ns DATASHEET_INPUT_SLEW 5.962569e-03ns DATASHEET_OUTPUT_SLEW 5.968207e-03ns DATASHEET_PARAMETER {{ VDDG2 0.1715pf 1.355e- 06mW} {VDDG 0.1715pf 3.62e-06mW} {VDD 0.1715pf 2.614e-06mW}} PEAK_CURRENT {{VDDG2 0.25mA {VDDG 0.2mA} {VDDG 0.15mA}} END_MODE
	

Support for Specifying the TSV Subckt File during Power-Grid Library Generation

Now, you can use the `-tsv_subckt_modelfile_list` parameter of the `set_pg_library_mode` command to specify the name of external Through Silicon Via (TSV) subckt model files to be used for 3D-IC extraction. You can specify a list of files separated by space. Each file contains the subckt model information for TSV. The specified file has the Spice format.

A sample file is shown below:

```
.subckt TSV top bottom
```

```
R1 top n2 0.03075
R2 n2 bottom 0.03075
C1 n2 0 28f
.ends
```

Use Model:

`set_pg_library_mode -tsv_subckt_modelfile_list` *`filename1 filename2 ...`*

Ability to Reuse Electrical Data for Macro PGV Generation

Macro PGV generation flow has been optimized to enable the reuse of electrical data from an old PGV to create a new PGV. The simulation steps are skipped for reused PGV flows, which results in reduced runtime.

To support this enhancement, two new parameters have been added to the `set_advanced_pg_library_mode` command:

- `-pgv_path`: Specifies the path of the cell library from which the electrical data will be used to create a new PGV.
- `-reuse_electrical_data`: When set to true, specifies to reuse the electrical data from the old PGV and only process the physical part in the new PGV. The default value is `false`.

Use Model:

`set_advanced_pg_library_mode -pgv_path` *`<path>`* `-reuse_electrical_data true`

Support for XDSPF and PGBD Inputs made Obsolete

The following parameters of `set_advanced_pg_library_mode` command are now obsolete. These parameters are no longer supported for PGV generation.

- `-import_xdspf_list_file`
- `-pgdb_layermap`
- `-pgdb_list_file`
- `-xdspf_layermap`

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Static and Dynamic Power Analysis

Release 26.10 Enhancements

Partition SPEF Support

The `power_partition_spef` attribute has been introduced to generate block-level SPEF files from a flat SPEF file. This enhancement allows you to process input SPEF data and automatically generate partition-level SPEF chunks. The `power_partition_spef` attribute is supported only in the extensively parallel (XP) mode for full-chip designs using flat SPEF files.

The use model of this feature is:

```
set_power_spef_files { flat.spef.gz }
```

```
set_db power_partition_spef true
```

Default Change for Unified Power Switch Flow in XP Flow

The default value for the `power_unified_power_switch_flow` attribute has been changed to true in the XP flow.

This change aligns the default behavior with the existing rail analysis parameter, `set_rail_analysis_config -unified_power_switch_flow`, which is already set to `true` by default in the XP flow, ensuring consistency across both power and rail analysis modes. Due to this default behavior change, you may observe differences in power and current analysis results compared to previous versions. If you need to maintain the previous behavior, explicitly set the `power_unified_power_switch_flow false` attribute in your scripts.

Enhanced Support for Logical Hierarchy and Domain-Based Activities

Voltus has been enhanced to support multiple activities per block. Previously, the tool honored only a single activity for each block. With this enhancement, you can now define different switching activities for different domains and/or logical hierarchies within the same DEF block during stable flop scheduling.

To enable this functionality, the following options have been introduced:

- `set_default_switching_activity -pg_net` <*netname*> - Specifies the name of the power net for domain-based switching activity assignment.
- `power_honor_default_switching_activity_in_scheduling {true|false}` - Controls whether default switching activity settings are applied during hierarchy/domain-based scheduling.

This feature improves accuracy for complex hierarchical designs with varying switching characteristics and optimizes flop scheduling based on realistic switching activity patterns.

Example:

Following is an example of domain-based activity specification in the dynamic vectorless flow:

```
set_default_switching_activity -sequential_activity 0.7 -input_activity 0.7
macro_activity 0.5 -clock_gates_output 2 -pg_net VDDM
```

Support Added for Ignoring Unconstrained Nets and Pins with Valid Timing Data

Some nets/pins lack timing constraints (no associated clock) but contain valid timing data with arrival windows in the TWF file. Previously, the tool applied default frequency toggles to these unconstrained nets, which could cause false IR violation warnings due to high voltage drop.

You can now use the `power_ignore_unconstraint_net_with_timing_data true` attribute to exclude unconstrained nets from toggle generation. This enhancement eliminates false IR violation reports and provides better control over toggle generation.

Scaling Capacitance by Slew Ratio

When using the lumped capacitor model, long net lines may cause reduction of effective capacitance (Ceff) due to high resistance shielding and low slew ratio. This can lead to overly optimistic delay estimate during Liberty lookups of lower load.

In this release, the following attributes have been added to the `power` category to adjust and scale the load capacitance using the slew ratio:

- `power_use_effective_load_cap_method {lumped | max | avg | min}` - Specifies the method to determine the slew ratio. This parameter has the following arguments:
    - `lumped` - the slew ratio is 1, which indicates that a lumped capacitance is used.
    - `max` - the slew ratio is calculated as max(slew of all drivers)/ max(slew of all receivers)
    - `min` - the slew ratio is calculated as min(slew of all drivers) / min(slew of all receivers)
    - `avg` - the slew ratio is calculated as avg(slew of all drivers) / avg(slew of all receivers)
- `power_effective_load_cap_slew_threshold_limit <float between 0 and 1>` - the minimum threshold value for slew ratio to prevent effective capacitance from being derated beyond realistic limits.

This enhancement eliminates over optimism and enables more accurate modeling for long nets.

Example:

```
set_db power_method dynamic_vectorless
set_db power_enable_state_propagation true
set_db power_enable_xp true
set_db power_use_effective_load_cap_method max
set_db power_effective_load_cap_slew_threshold_limit 0.9
```

New Attribute to Force Propagation

In the dynamic vectorless flow, Voltus performs logic simulation by generating the state of an instance's output pins based on the states of its input pins. Occasionally, the output may become fixed or constant. In these scenarios, you can use the `power_enable_force_propagation true power` category attribute to ensure that the output toggles are according to the clock signal rather than the input pin states. This attribute assigns clock activity to outputs when input propagation is not feasible, thereby enhancing overall activity.

off_pg_nets Parameter Enhanced to Support Reporting of Bulk Pin Leakage

The `power_off_pg_bulk_leakage` power attribute has been introduced to account for the leakage power through an instance's bulk pin when it is connected to a power-ground (PG) rail in the ON state.

The following diagram illustrates that the instance contributes bulk pin leakage to the ON state rail (**vdd**) through the *VBP* pin and zero leakage to the OFF state rail (`vddlsw`) through the VDD pin.

```
 ----- vdd (AON) ---------------+------------
                                |
                                |  +------+
                                |  |      |
                                +--+VBP   |
                                   |      |
                                   |      |
                                +--+VDD   |
                                |  |      |
                                |  +------+
                                |
 ------ vddlsw (off_pg_net) ----+------------
```

Support Added for Ignoring Unconstrained Nets and Pins with Valid Timing Data

Some nets/pins lack timing constraints (no associated clock) but contain valid timing data with arrival windows in the TWF file. Previously, the tool applied default frequency toggles to these unconstrained nets, which could cause false IR violation warnings due to high voltage drop.

You can now use the `power_ignore_unconstraint_net_with_timing_data` true attribute to exclude unconstrained nets from toggle generation. This enhancement eliminates false IR violation reports and provides better control over toggle generation.

Custom Output Directory for Thermal Results

You can now use the `set_thermal_analysis_mode -thermal_output_dir` parameter to specify a custom output directory for thermal analysis results instead of using the default `ThermalDir`. This enhancement gives you better control over where your output files are stored, making it easier to manage results across multiple runs.

Example:

Use the `-thermal_output_dir` parameter to define your preferred directory path:

```
set_thermal_analysis_mode -tool_path $CELSIUSTOOL \
                          -boundary_condition_file bc_75.txt \
                          -temperature_map_file temperatureMap_75 \
                          -thermal_output_dir thermal_dir_new \
                          -power_map_file vtm_output/top.vtm.txt

report_thermal
```

Instance-Based FIT Reporting with New Parameters

In this release, two new parameters are added to the `report_power` command to generate instance-level FIT report based on cell-level data:

- `-cell_fit_file` *`filename`* - Specifies a file that accepts a cell-based FIT input table with two columns: cell name and FIT value.
- `-output_cell_fit_file` *`filename`* – Specifies the name of the output file for the generated FIT report, which includes three columns: instance name, corresponding cell name, and computed FIT value.

This enhancement ensures greater visibility into instance-level risks and supports more efficient debug and review of high-FIT contributors. Instance-based FIT report is supported only in the XP mode.

New Attribute to View Cell Usage Statistics

The `power_show_cell_usage_stats {true | false}` attribute has been added to the `power` category. It provides cell usage statistics alongside power values, indicating the number of cells used from a specific library and the corresponding power percentage. This helps identify libraries with the greatest and least impact on power consumption.

You must ensure that this attribute is specified before using the `report_power` command, as demonstrated below:

```
set_db power_show_cell_usage_stats true
report_power -out_file p1.rpt
```

Following is a snippet from the report:

```
Cell usage statistics:
Library j1os_h , 88 cells ( 7.351713%) , 6.56434e-06 mW ( 0.554089%  )
Library j1os_l , 123 cells ( 10.275689%) , 0.000799809 mW ( 67.511081%  )
Library j1os , 986 cells ( 82.372597%) , 0.000378334 mW ( 31.934830%  )
```

The `power_show_cell_usage_stats` attribute is applicable only to the static power calculation, Verilog-based (`power_static_netlist verilog`) flow.

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Rail Analysis

- Ability to Reuse HPT Data
- set_die_model Command Enhanced

Release 26.10 Enhancements

Enhanced Rail Analysis Reporting with Average Effective Resistance

A new argument, `avg`, has been added to the `set_rail_analysis_config -r_effective_pin_report_method` parameter to improve reporting reliability for signoff analysis. When analyzing large macro cells, instances at different physical locations can exhibit varying effective resistance (reff) values. These location-dependent variations can introduce instability in signoff results, making it challenging to achieve consistent and reliable analysis outcomes.

The `avg` argument addresses this challenge by providing averaged effective resistance values, which offer greater stability compared to individual node measurements.

Usage Models

- `-r_effective_pin_report_method avg` - Reports only the average effective resistance value for instance pins with multiple nodes
- `-r_effective_pin_report_method {worst avg best}` - Reports comprehensive effective resistance data: best, worst, and average values for instance pins with multiple nodes

Generates Multiple RLRP and REFF Reports in a Single Run

The `set_rail_analysis_config` command has been enhanced with improved `-r_effective_pin_report_method` and `-rlrp_pin_report_method` parameters that now support generating multiple report types simultaneously. This enhancement accelerates closure of RLRP and REFF issues by having all necessary report variants readily available for comparison and analysis. Each report type is automatically saved to a separate file, making it easy to access specific analysis results.

Example:

```
set_rail_analysis_config -r_effective_pin_report_method { best avg worst}
```

Output files:

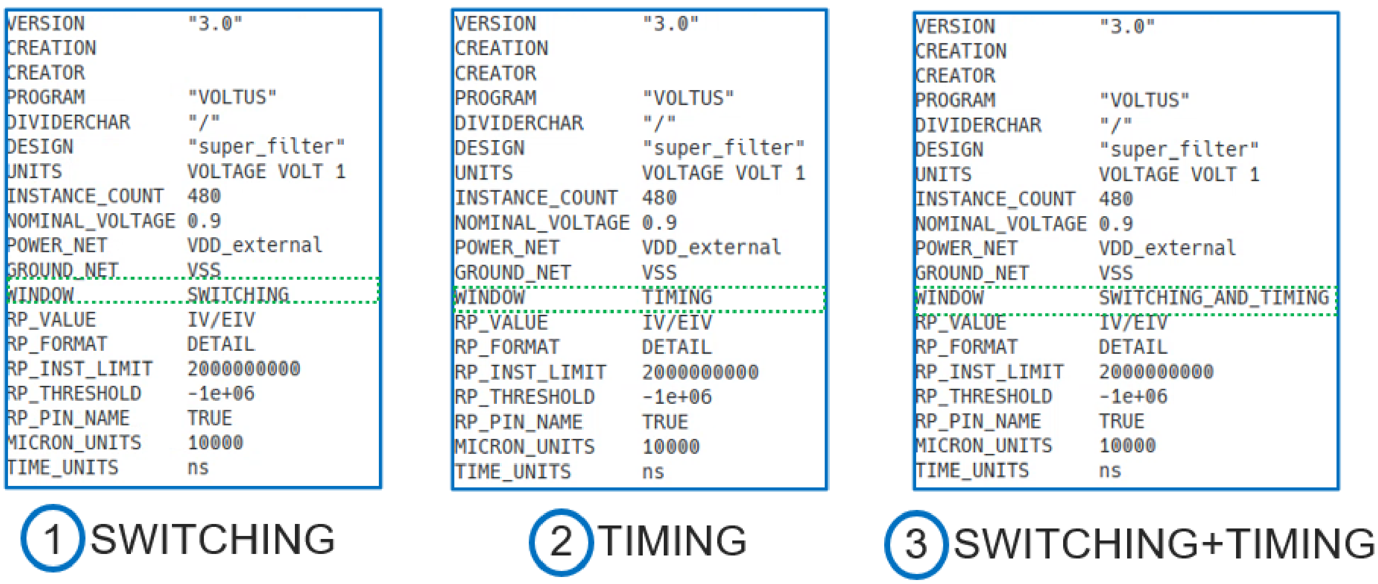
```
net.best.effr
net.avg.effr
net.worst.effr
domain_effr.best.rpt
domain_effr.avg.rpt
domain_effr.worst.rpt
```

Separate Timing and Switching Window EIV Reports

The `set_rail_analysis_config` `-eiv_eval_window` parameter has been enhanced to provide greater flexibility in EIV report generation during rail analysis. Rail analysis reporting now supports generating separate EIV reports for both timing and switching windows within a single analysis run. Previously, you could use the `both` argument to generate a single combined report containing both switching and timing-window EIV data. Now, you can use the `-eiv_eval_window {switching timing}` parameter to generate distinct reports for each window type in a single run. This enhancement allows easier post-processing and review of specific EIV analysis results, generating both report types simultaneously without additional runs.

Example:

The following image shows the EIV reports for the switching window, timing window, and both:



The type of report generated is based on the following use models:

- `-eiv_eval_window {both}` - Reports 3 Only (*Default Behavior)
- `-eiv_eval_window {switching}` - Reports 1 Only
- `-eiv_eval_window {timing}` - Reports 2 Only
- `-eiv_eval_window {switching timing}` - Reports 1 and 2
- `-eiv_eval_window {switching both}` - Reports 1 and 3

When multiple EIV methods (best and worstavg) are enabled, the file naming convention for the EIV reports is based on the EIV method and EIV evaluation window, as shown in the following examples:

```
VDD_VSS.worstavg.timing.iv
VDD_VSS.worstavg.switching.iv
VDD_VSS.best.timing.iv
```

New Parameter to Specify the Lifetime Rating Factor

A new attribute, `check_ac_limit_lifetime_rating_factor_file`, and a new parameter, `set_rail_analysis_config -lifetime_rating_factor_file`, has been introduced to specify a lifetime rating factor file that contains jmax_life values for each layer. These values are defined based on temperature and lifetime pairs. When you provide this file, EM analysis will automatically select the appropriate `jmax_life` factor for each layer according to the specific temperature-lifetime combinations defined within the file. The lifetime and EM temperature can be specified using the `set_rail_analysis_config -lifetime` *`hours`* and `-em_temperature` *`°C`* parameters.

Following is a snippet of the lifetime rating factor file:

```
*Rating factor of M0/V0/M1/V1/M2/V3/M3
TEMPERATURE         90       95       100      105      110 ...
LifeTime

----------------------------------------------------------------------------------------
0.5                 2.578    2.242    2.024    1.774    1.547
1                   2.520    2.191    1.979    1.734    1.512
2                   2.430    2.113    1.908    1.672    1.458
3                   2.382    2.072    1.871    1.639    1.430
...

----------------------------------------------------------------------------------------
```

Voltus InsightAI Command Enhanced

The `set_voltus_diagnostics_mode` command now includes new parameters to support enhanced scenario-based configuration (differing EIV method, EIV evaluation window, and thresholds). These parameters allow scenarios to be defined for various groups or collections within the same repair/report session, enabling greater precision and flexibility. Instead of using uniform threshold or EIV values across all instances, you can customize settings based on:

- Instance collections
- Master blocks

- Rail database directories

Example: If the general clock threshold is set to 10% and the data threshold to 12%, a critical instance group can have a stricter 9% threshold. All three threshold settings are honored in the same session. This allows multiple scenarios with different threshold configurations to run simultaneously in a single session.

New Parameters	Description
`-disable_layer` *`filename`*	

	Specifies the name of the file that contains the list of layers to be disabled or excluded during pre-route and post-route RPG.
	The format of the file is:
	`Layer1`
	`Layer2`
	`Layer3`

`-instance_mapping` *`filename`*	

	Specifies the name of the file that contains the mapping between a scenario and its corresponding instance files. Each scenario has its own set of instance files. Each line in an instance file defines one instance name or block instance name.
	
	Following is a snippet of the instance mapping file (`instance_mapping.txt`):
	
	`SCENARIO_1 {instance1.txt instance2.txt}`
	`SCENARIO_2 {instance1.txt}`
	
	The content of the 2 instance files is shown below:
	
	`instance1.txt`
	`------------`
	`Inst_A*`
	`Inst_B*`
	
	`instance2.txt`
	`-------------`
	`Inst*`

`-master_list {`*`list_of_master_blocks`*`}`	
	Specifies the list of master blocks to be reported for IR repair.
`-master_mapping` *`filename`*	

	Specifies the name of the file that contains the mapping between a scenario and its corresponding master files. Each scenario has its own set of master files. Each line in a master file defines one master block.
	
	Following is a snippet of the master mapping file (`master_mapping.txt`):
	
	`SCENARIO_1 {master1.txt master2.txt}`
	`SCENARIO_2 {master1.txt}`
	
	The content of the 2 master files is shown below:
	
	`master1.txt`
	`-------------`
	`Top`
	
	`master2.txt`
	`-------------`
	`Super_filter`

Enhanced Parameter	**Description**
`-db_mapping` *`filename`*	

	Specifies the name of the file that contains the mapping between a scenario and its corresponding rail directories. Each scenario can be mapped to multiple rail directories. Previously, one scenario could be mapped to only one rail directory.
	
	Following is a snippet of the rail database mapping file (`db_mapping.txt`):
	
	`SCENARIO_1 {RailDB1 RailDB2}`
	`SCENARIO_2 {RailDB1}`

New Parameter for Violation Magnitude-Based Repair with External Signoff Tool Consistency

A new parameter, `set_voltus_diagnostics_mode -violation_input` *`filename`*, has been introduced, enabling repair operations based on violation specifications that align with an external signoff tool. This feature allows Voltus to replicate violation magnitudes for individual instances, ensuring consistent repair behavior between the external signoff tool and Voltus while eliminating the need for global threshold adjustments.

You can specify a text file containing a per-instance violation specification in the following format:

```
instance_name   threshold(mV)   violating_amount(mV)
Example:
inst_1  100  35
inst_2  80  20
```

For instance 1 (inst_1), an external signoff tool detected a 35mV violation against a 100mV threshold. This violation data allows Voltus to perform repairs using the same violation magnitude values reported by the external tool, ensuring consistency between the tools.

SHE Parameter File Specification

A new parameter, `-parameter_file`, has been introduced for both the `set_self_heat_analysis_mode` and `report_self_heat` commands. This parameter allows you to specify the self-heating effect (SHE) analysis parameter file name (param.sh) required for SHE calculations. This foundry-provided file contains essential alpha and beta coupling factors that the tool uses for coupledT calculations when modeling heat transfer from devices to metal layers.

Partial Net Coverage Reporting in SPEF

A new `check_ac_limit` category attribute, `check_ac_limit_report_unannotated_shapes {true | false}`, has been introduced to enhance visibility into partial net coverage within SPEF files. When this attribute is set to `true`, the `<`*`report_name`*`>.rpt.unannotated_shapes` is generated. This report identifies and lists nets that contain shapes without resistance annotations, or unanalyzed/ partially analyzed segments. The feature helps to detect incomplete parasitic extraction coverage and improve accuracy during validation and signoff workflows. The reporting of partial net coverage works only when `check_ac_limit_effort_level` is set to `high`.

Use Model:

```
set_db check_ac_limit_effort_level high check_ac_limit_report_unannotated_shapes true
```

Report Format:

```
NET: <net_name>          <- Specifies the name of the analyzed simulation nets with
```

```
unannotated shapes
LAYER (x1 y1 x2 y2)      <- Layer name and shape coordinates
```

Sample Report:

```
NET: Net_A
LAYER (x1          y1          x2          y2)

M2    (30.2850    12.0800    37.6150    12.2200)
M2    (30.2000    12.0650    30.2850    12.2350)
...
```

This attribute applies only when running in DP mode; it is not supported in XP mode.

Ability to Compress Rail Output Files

A new parameter, `-report_compress_level`, has been added to the `set_rail_analysis_config` command to compress the rail output files in the GNU Zip format. The compressed files will have the suffix `.gz`.

Use Model:

```
set_rail_analysis_config -report_compress_level {0 | 1-9}
```

- `0`: default, no compression
- `1-9`: the compression levels vary from 1 to 9. 1 provides the fastest compression speed but at a lower ratio, so the file size will be larger. 9 provides the highest compression ratio but at a lower speed.

set_power_data Command Enhanced

The `-cell` parameter has been added to the `set_power_data` command to specify power from the block-level run for the multiple instances of the same block. Therefore, you no longer need to specify power for each instance separately. This option is useful when power is calculated for hierarchical partitions of the design.

Use model:

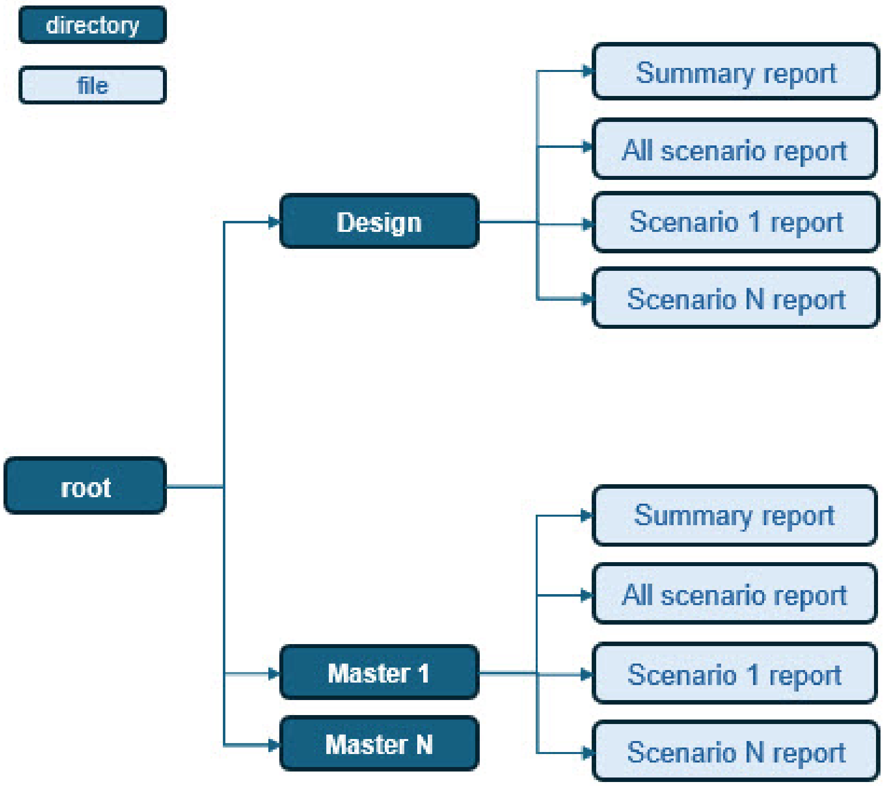
```
set_power_data -cell cellname
```

Example:

```
set_power_data -cell BlockA -format current {file1, file2, file3 ...}
```

Voltus InsightAI Reporting Capabilities

A new REPORTS directory has been added at the root level to better organize the Voltus InsightAI analysis results. The following diagram illustrates the directory structure:



Here, the number of subdirectories created depends on the hierarchical structure of the design. The REPORTS directory includes the following:

- The Design directory contains the reports for the complete design.
- Each hierarchical master, including the top level, has its own directory that contains the corresponding reports.

Example
For a design with master blocks A and B, the following four directories are created:

- Design – full design reports
- Top Block – top-level reports
- Master Block A – reports for master block A
- Master Block B – reports for master block B

To support this enhancement, the following parameters have been introduced:

set_voltus_diagnostics_mode	
-all_scenario_ranking {wiv | tiv}	Specifies that all scenarios are ranked either based on the total number of violations (TIV) or worst violations (WIV).
-scenario_ranking {wiv | tiv |powerrlrp | powerreff | groundrlrp | groundreff}	
	Specifies that the scenario ranking is done for a single scenario.
-ground_reff_threshold *value*	Specifies that the violation report contains only instances that have values less than the specified ground REFF threshold.
-ground_rlrp_threshold *value*	Specifies that the violation report contains only instances that have values less than the specified ground RLRP threshold.
-power_reff_threshold *value*	Specifies that the violation report contains only instances that have values less than the specified power REFF threshold.
-power_rlrp_threshold *value*	Specifies that the violation report contains only instances that have values less than the specified ground RLRP threshold.
get_ir_diagnostics	
-report_mode {summary_only | detailed}	Controls the type of reports generated.

Example

```
set_voltus_diagnostics_mode -config config.txt -domain ALL -db_mapping db_mapping.txt
get_ir_diagnostics -report_mode detailed
```

Enhanced Voltus InsightAI Commands for Timing Path Protection during PG Reinforcement

The `set_voltus_diagnostics_mode` and `get_voltus_diagnostics_mode` commands now include new parameters to help protect timing paths during PG stripe reinforcement.

set_voltus_diagnostics_mode

`-path_group {ALL | reg2reg}`	
	Specifies which timing paths to protect during reinforce_pg.
`-timing_protection_config` *`filename`*	
	Defines timing degradation thresholds per timing view using a configuration file.
`-initialize`	
	Saves the current power settings before running the Insight repair commands using internal power/rail values. After InsightAI-based analysis, this parameter enables restoration of the original power settings.
`get_voltus_diagnostics_mode`	
`-timing_protection_config` *`filename`*	
	Specifies to automatically generate the timing protection configuration file. This file includes the timing degradation thresholds per timing view.

Generation of Voltus InsightAI Database

Voltus has been enhanced to generate the InsightAI database from rail analysis results. This database can be directly used for subsequent IR drop fixing operations, eliminating the need to reload the rail results. In addition, the database preserves the reinforce power-grid information, which is then applied to support incremental power-grid fixing. To enable this workflow, two new parameters have been added to the `set_voltus_diagnostics_mode` command: **`input_insight_db`** and **`output_insight_db`**.

Use Model

1. Generate the output database using the `-output_insight_db` parameter. The database is stored in the path: <output_db>/master_directory).
   The indicative structure of the database is as following:
   ```
   <output_db>/master1/
   <output_db>/master2/
   <output_db>/master3/
   ```

2. Use the generated database as the input database specified with `-input_insight_db`. The specified database will be one of the master databases generated using `-output_insight_db`. You can then run `add_reinforce_pg` without loading the entire rail database, enabling you to fix issues without reloading the rail result.

Example:

1st Step: Output Database Generation

```
set_voltus_diagnostics_mode -config <insight_config> -import_analysis_data -db_mapping
<db_mapping_file> -output_insight_db <output_db>
get_ir_diagnostics -mode repair_directive_only | report_and_repair_directive
```

The database is generated under the directory <*output_db*>

2nd Step: Insight Database Fixing

```
set_voltus_diagnostics_mode -config -input_insight_db <output_db>
add_reinforce_pg -auto_ir_fix postroute
```

write_die_model Command Enhanced

The `-local_reference_ground` parameter has been added to the `write_die_model` command to assign a ground source as reference ground locally for sources in the current file `.isrc`. You can specify the reference node in the PLOC file as "`RefGnd=PortName`" to compute the isrc of port by considering reference ground locally during die-model generation.

Use model:

```
write_die_model \
-local_reference_ground true \
-port_grouping_by_ploc_files true \
-generate_SIPI_port_definition_file true \
-disable_isrc_node_zero true \ (make isrc with ref ground)
-disable_global_node_zero false (keep global node zero in ckt)
```

write_current_region Command Enhanced

A new parameter, `-on_interdie_bump`, has been added to the `write_current_region` command to create tap nodes only on non-vsrc PLOC nodes. When the option is set to `true`, the number of total tap changes to include the tap nodes only on non-vsrc PLOC. You can check the change in total taps in the “Grid Statistics of Net XXX” section of the log file at `<state directory>/logs/design_report/dmg/ext_report.log`

Use Model:

```
write_current_region -on_interdie_bump true
```

Runtime Optimization for Reading the DEF files

Voltus has been enhanced to parse the largest DEF file on the master machine. You can then reuse the data of the parsed-DEF file from the specified directory using the `-xp_reuse_parse_def_directory` parameter of the `set_rail_analysis_config` command in the XP mode. This enhancement significantly reduces the runtime required for reading the large DEF files.

Use Model:
`set_rail_analysis_config -xp_reuse_parse_def_directory` *`directory_name`*

Example:
```
set_rail_analysis_config -xp_reuse_parse_def_directory
/DefHierReport/rail_dynamic/core_125C_dynamic_1
```

Support to Suppress the Negative RC values

A new parameter, `-suppress_negative_rc`, has been added to the `write_die_model` command to suppress the negative resistor and capacitor (RC) values during die-model generation. When this parameter is set to `true`, the generated netlist contains positive RC values with voltage-controlled (VC), current-controlled(CC), and function-controlled (FC) sources to ensure accuracy and stability in the die models.

Use model:
```
write_die_model -suppress_negative_rc true
```

Voltus InsightAI Enhancements

The following new parameters have been added to the `get_ir_diagnostics` commands to identify aggressor candidates with maximum switching currents to guide power integrity improvements:

Parameter	Description
`-cap_lower_limit`	Specifies the minimum load capacitance value driven by aggressor candidates in the `eco_report` mode.
`-cap_upper_limit`	Specifies the maximum load capacitance value driven by aggressor candidates in the `eco_report` mode.

read_power_rail_results Command Enhanced

Starting this release, you can use the `-force` option of the `read_power_rail_results` command to block the pop-up window that appears when the `read_power_rail_results -reset` parameter is specified. The pop-up window displays a confirmation message for resetting the power and rail directories.

Use Model:

```
read_power_rail_results -reset -force
```

Single Partition Support During Extraction

Voltus now supports single partitioning when extracting the specified metal layers. This helps to minimize extraction fracturing at partition boundaries, leading to better RC grid quality. With this enhancement, the false EM violations on the partition boundaries can be avoided.

To support this, two new parameters are added to the `set_rail_analysis_config` command:

- `-single_partition_extraction_layer` - Specifies to extract RC using a single partition on a top metal layer.

  Use model:

  ```
  set_rail_analysis_config -single_partition_extraction_layer {METAL_LAYER }
  ```

  Example:

  ```
  set_rail_analysis_config -single_partition_extraction_layer {METAL_8}
  ```

- `-single_partition_extraction_layer_list` - Specifies to extract RC using a single partition on multiple layers.

  Use model:

  ```
  set_rail_analysis_config -single_partition_extraction_layer_list {METAL_LAYER_1
  METAL_LAYER_2 …… }
  ```

  Example:

  ```
  set_rail_analysis_config -single_partition_extraction_layer_list  {METAL_5
  METAL_8}
  ```

Ability to Exclude Instances during Rail Extraction

A new parameter, `-inst_ignore_file`, has been added to the `set_rail_analysis_config` command to exclude the LEAF instances mentioned in the specified file during rail extraction.

Use Model:

```
set_rail_analysis_config -inst_ignore_file {filename}
```

File Format:

```
Inst1
Inst2
.
.
.
InstN
```

Support to Read Power Data to Identify PTI Files

The `-power_directory` parameter has been added to the `set_power_data` command to read the power data from the specified directory for cell power. You no longer need to define each PTI file individually. Instead, you can use this parameter to read `power_results.json` and automatically identify all the PTI files. If the `power_results.json` is not found in the specified directory, then the power data could be retrieved from `dir/voltus_power_output.latest`. When specified, the `-power_directory` parameter takes precedence and bypasses the `-format` parameter.

Use Model:

```
set_power_data -cell <cell name> -power_directory <power directory for cell power>
```

Example:

```
set_power_data -cell top_design -power_directory ./power
```

Ability to Reuse HPT Data

A new parameter, `-xp_reuse_hpt_directory`, has been added to the `set_rail_analysis_config` command to support the reuse of hierarchical path tracing (HPT) data from the HPT directory of a previous run with the modified extraction options. With this enhancement, you no longer need to run HPT again if there are no DEF changes in the component section or logical connectivity and can use the earlier HPT data to avoid unnecessary runtime. The reuse of HPT data is also allowed if the DEF changes are related to routing shape only.

```
set_rail_analysis_config -xp_reuse_hpt_directory <state directory>
```

set_die_model Command Enhanced

A new parameter `-static_current` has been added to the `set_die_model` command to use the average of PWL waveform as the static current for static rail analysis. Earlier, only the current value at `t=0` from the PWL waveform was considered for static run.

Use Model:

```
set_die_model -static_current initial | average
```

Example:

```
set_die_model -static_current average
```

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Multi-Die IR Analysis

Release 26.10 Enhancements

Voltus IR Co-simulation with Multi-die and Package

Advanced 2.5D and 3D packaging technologies (InFO, CoWoS, WoW) are increasingly complex for power integrity analysis due to heterogeneous chip integration and high-performance requirements, with hundreds of millions of instances. Voltus multi-die capability in XP mode delivers comprehensive chip-package-board power signoff co-analysis, enabling:

- Concurrent static/dynamic IR/EM analysis across dies and silicon interposers
- Accelerated runtime performance
- Net- and pin-based package specification for HBM die-models and stacked dies
- Full system PG connectivity and IR/EM verification through co-simulation

Note: Multi-die analysis is available exclusively in XP mode.

For more information, refer to Voltus Multi-die IR Analysis.